OSINT and Data Mining Analysis: Extracting TTB Certificate of Label Approval (COLA) Records and Compliance Rejections

The Federal Alcohol Administration (FAA) Act establishes a rigorous regulatory framework dictating the packaging, labeling, and advertising of alcoholic beverages introduced into interstate commerce within the United States. Under this mandate, the Alcohol and Tobacco Tax and Trade Bureau (TTB), a bureau of the Department of the Treasury, requires domestic producers, bottlers, and importers to obtain a Certificate of Label Approval (COLA) via TTB Form 5100.31 prior to the commercial release of any wine, malt beverage, or distilled spirit.[1] The TTB processes an excess of 100,000 label applications annually, generating a massive and continuously updating repository of regulatory data.[3] Within this expansive data ecosystem, the identification, extraction, and analysis of non-compliant, "Needs Correction," and strictly "Rejected" applications present a highly complex Open-Source Intelligence (OSINT) and data mining challenge.

The fundamental difficulty in acquiring rejected COLA datasets stems from the TTB's strict data visibility protocols. The agency intentionally obscures failed or pending applications from its public-facing databases to protect industry confidentiality and intellectual property during the product development phase.[4] Consequently, analysts, compliance developers, and market researchers cannot rely on a single, centralized endpoint to harvest data on compliance failures. Instead, the extraction of this data necessitates the deployment of advanced data mining techniques, alternative repository scraping, deep analysis of industry compliance graveyards, and the exploitation of Freedom of Information Act (FOIA) reading room logs to reconstruct the precise taxonomies of regulatory failure.

The ensuing report delivers an exhaustive, multi-vector technical analysis of the TTB COLA system architecture. It provides actionable extraction methodologies, explicit query parameters, dataset locations, and programmatic workflows required to harvest both approved label data and the elusive repositories of rejected alcohol label applications.

Part 1: Architecture of the TTB COLA System and the Application Lifecycle

To execute effective data extraction protocols against the TTB COLA database, an investigator must first map the digital architecture of the COLAs Online system and understand the precise lifecycle of an application. The system operates on a bifurcated architecture: a secure, authenticated portal utilized by industry members and third-party filers for the submission of applications, and a restricted, unauthenticated Public COLA Registry designed for consumer

oversight, competitor analysis, and state-level compliance verification.[5]

The 14-Digit TTB ID Syntax and Metadata

Every application submitted to the TTB, whether electronically through the COLAs Online portal or via physical paper submission, is assigned a unique 14-digit identifier known as the TTB ID. This alphanumeric string serves as the primary key for all database queries and is essential for targeted extraction operations. Understanding the morphology of the TTB ID is critical for programmatic scraping, data validation, and temporal sorting. The first five digits of the TTB ID contain embedded temporal metadata representing the exact date the application was received by the agency.[7] Specifically, the first two digits represent the calendar year, and the subsequent three digits represent the Julian date.[7] For example, a TTB ID beginning with the sequence 24045 unequivocally indicates an application that was ingested on the 45th day of the year 2024. The remaining nine digits constitute an internal sequential routing number and commodity identifier assigned by the COLAs Online database architecture.[7]

The Application Status Continuum and Data Visibility

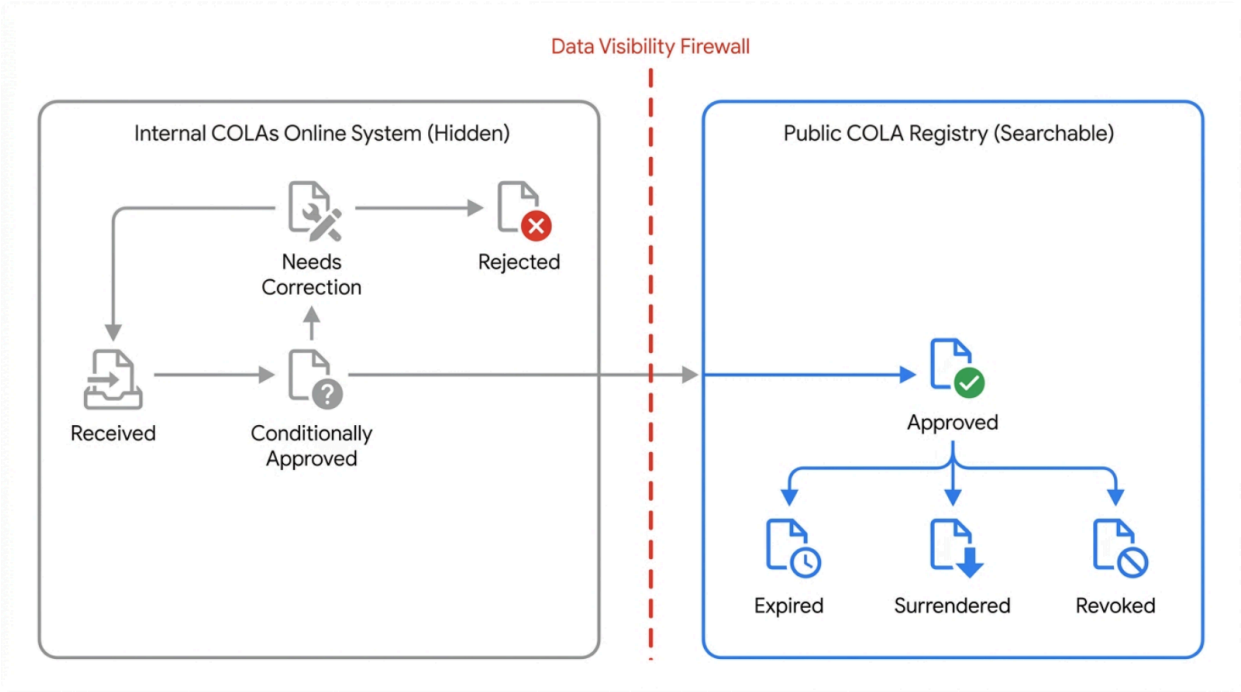
The COLAs Online system tracks applications through a highly specific set of internal statuses. Recognizing these operational states is vital for OSINT investigators, as the current status of an application strictly dictates whether its associated data payload is publicly exposed or hidden behind the authenticated firewall. The application continuum proceeds through the following potential states:

1. **Received:** The application has been ingested into the system, assigned a TTB ID, but has not yet been assigned to or reviewed by a specialist within the Alcohol Labeling and Formulation Division (ALFD).[8] At this stage, the data is entirely confidential.
2. **Conditionally Approved:** Introduced to streamline the regulatory pipeline, this intermediate status indicates that an ALFD specialist has reviewed the application and proposed minor administrative changes.[9] These changes typically involve aligning the data entered into the digital form with the visual information present on the uploaded label image (e.g., modifying a "Brand Name," "Fanciful Name," "Grape Varietal," or "Appellation" field).[9] Upon receiving this status, the applicant is granted a strict seven-day window to accept or decline the specialist's proposed changes.[10] If the changes are accepted, the application achieves immediate approval. If declined, or if no action is taken within the seven-day timeframe, the system automatically degrades the application to "Needs Correction".[10] This status is restricted to the authenticated portal.
3. **Needs Correction:** This is a critical intermediate status and a primary target for compliance analysis. Approximately 43% of all label applications submitted to the TTB are returned to the submitter under the "Needs Correction" status at least once during their lifecycle.[11] The application is returned with specific annotations from the reviewing specialist detailing the exact regulatory violations or formatting errors present on the label or within the application fields.[4] Upon entering this status, the submitter is afforded exactly 30 calendar days to rectify the identified errors and resubmit the application without losing

their priority placement in the processing queue.[7] Applications in the "Needs Correction" state are strictly hidden from the public and are completely inaccessible via the Public COLA Registry.[6]

4. **Rejected:** This represents the terminal state of a failed application. An application is formally marked "Rejected" if the submitter fails to execute the required corrections and resubmit within the allotted 30-day window, or if the application is resubmitted but still fails to resolve the original compliance issues.[4] Furthermore, if the fundamental premise of the label represents an uncorrectable violation of the FAA Act, it may face outright rejection. Crucially, rejected applications are strictly and permanently excluded from the Public COLA Registry.[5]
5. **Approved / Certificate:** This is the terminal state of a successful application. The label is granted a Certificate of Label Approval, legally authorizing the product for interstate commerce. The associated data and high-resolution label images are pushed across the firewall to the Public COLA Registry, generally becoming viewable to the public within 48 hours of final approval.[12]
6. **Surrendered / Revoked:** These are secondary statuses applied post-approval. A COLA may be voluntarily "Surrendered" by the permit holder, often due to trademark disputes, regional association protests, or formula changes. Conversely, a COLA may be forcibly "Revoked" by the TTB following market audits, product integrity investigations, or the discovery of illicit advertising claims.[5] Both of these statuses remain publicly visible within the registry.[5]

The TTB COLA Visibility Firewall Restricts Public Access to Failed Applications



Applications marked as 'Needs Correction' or 'Rejected' remain securely behind the TTB's authenticated portal. Only Approved, Surrendered, Expired, and Revoked records cross the data firewall into the Public COLA Registry.

Part 2: Exploiting the TTB Public COLA Registry for Post-Market Anomalies

While the TTB explicitly states that "Rejected or returned applications are not included in this database" [5], mastering the search architecture of the Public COLA Registry remains an essential foundational step for compliance analysts. It allows investigators to extract "Surrendered" and "Revoked" labels—which serve as functional equivalents to post-market rejections—and establishes the baseline framework for automated scraping operations.

URL Endpoints, Query Strings, and Parameter Limitations

The primary portal for accessing the TTB Public COLA Registry is located at the following exact URL: https://www.ttb.gov/regulated-commodities/labeling/cola-public-registry [12]
The registry operates without requiring an API key, username, or password authentication, making its endpoints highly susceptible to systematic, programmatic scraping.[5] For direct extraction of specific, known COLA records, analysts can bypass the search interface entirely by utilizing the advanced display endpoint and manipulating the query string with the ttbid

parameter. The exact URL structure for direct record access is: https://www.ttbonline.gov/colasonline/viewColaDetails.do?action=publicDisplaySearchBasic&ttbid= [14]
Replacing the bracketed variable with a valid 14-digit TTB ID forces the system to render the complete HTML detail page for that specific application, exposing the submitted form data alongside high-resolution, printable versions of the approved label imagery.[14] The registry provides full electronic imagery for COLAs issued from 1999 to the present, while records pre-dating 1999 are restricted to text-based data only.[12]

Advanced Search and Filtering Methodologies

For bulk data extraction, the registry features an Advanced Search interface governed by a specific set of parameters and constraints. Analysts can manipulate these fields to isolate highly specific cohorts of labeling data.[15]

Search Parameter	Technical Description and Scraping Utility
Basic Permit / Brewer's No.	The most efficient parameter for scraping a single corporate entity's entire product catalog. Domestic wineries utilize a unique prefix, generally BW or BWN (Bonded Winery), followed by their specific registry number.[5]
Date Completed	Accepts a specific temporal range formatted strictly as MM/DD/YYYY. Programmatic scrapers must account for a hardcoded system limitation: the database restricts any single date-range query to a maximum span of 15 years.[15]
Wildcard Operator (%)	The system native SQL-style wildcard character is the % symbol.[5] This is an indispensable tool for OSINT operations relying on partial text strings or OCR outputs. For example, executing a query for %Valerie% will successfully return variations such as "Wines by Valerie," "Valerie's Vineyard," and "Chateau Valerie," effectively bypassing the limitations of exact-match string syntax.[5]
Type of Product	Allows filtering across the three primary regulated commodities: Wine, Distilled Spirit, and Malt Beverage.[15]
Type of Application	Distinguishes between standard interstate Certificate of Label Approval filings and domestic, intrastate Certificate of Exemption

	filings.[15]

When designing automated scrapers for this endpoint, developers must mitigate a rigid pagination constraint: the TTB caps all search results at a maximum of 500 items per query.[17] To harvest datasets exceeding this limit, scripts must be engineered to iterate through granular date segments or highly specific class/type combinations. Furthermore, the web interface provides a native "Save Search Results To File" link, which bypasses HTML parsing by directly generating a comma-separated values (.csv) payload containing top-level application metadata, including the TTB ID, Brand Name, Class/Type, and Origin Code.[17]

The "Status" Loophole: Extracting Pseudo-Rejections

Because the system architecture actively suppresses records holding the "Rejected" or "Needs Correction" status from the search dropdown menus, direct querying for these failures is technically impossible via this interface.[15] However, analysts can exploit the Status parameter by specifically targeting post-market compliance failures.

By executing a query with the Status parameter explicitly restricted to "Surrendered" and "Revoked," investigators can trigger a bulk dump of labels that initially bypassed the ALFD review process but subsequently collapsed under the weight of regulatory scrutiny, legal challenges, or field audits.[5] These records represent high-value anomalies: they are real-world examples of non-compliant labels that briefly entered the commercial stream before failing. Extracting this specific cross-section of data provides analysts with a foundational dataset of complex compliance errors that human specialists failed to detect during the initial submission phase.

Part 3: Compliance Graveyards and Industry Post-Mortems

With raw "Rejected" applications sequestered behind the TTB firewall, OSINT methodologies dictate a pivot toward secondary and tertiary sources. Legal blogs, compliance consulting databases, and specialized industry trade forums function as extensive "compliance graveyards." Within these domains, the exact TTB IDs, high-resolution physical label images, and the verbatim textual transcripts of TTB rejection notices are publicly archived, debated, and dissected by legal professionals and frustrated producers.

Legal Blogs and Case Studies: BevLaw and ShipCompliant

Law firms and software providers specializing in alcohol beverage compliance frequently publish granular case studies analyzing specific TTB rejections to educate their clientele. The most prominent and data-rich repositories for these post-mortems are the *Bevlog* (maintained by Lehrman Beverage Law) and the whitepapers generated by *Sovos ShipCompliant*.

The "Statement of Composition" (SOC) Rejection Typology: The BevLaw blog (bevlaw.com/bevlog/) routinely archives side-by-side visual examples detailing "what was submitted" against "why it was rejected." A prime target for data extraction involves Distilled Spirits Specialty (DSS) products. Under TTB regulations, products that do not meet a standard

of identity must feature a highly specific "Statement of Composition" (SOC) on the front label, accurately reflecting the approved formula.[18]

For example, if the TTB laboratory approves a formula dictating the SOC must read precisely *Liqueur, Whiskey, Caramel Color*, any deviation on the commercial label will trigger a rejection. In one documented case study, the producer attempted to market a product with the label text reading *A Seasonal Blend of Apple Cider Jack Daniel's Tennessee* while entirely omitting the required "Caramel Color" declaration.[18] The application immediately triggered a "Needs Correction" action.[18] Analysts mining the Bevlog archives can extract the exact parameters of these SOC disputes, noting that even microscopic deviations—such as altering the approved phrase "with natural flavor" to the plural "with natural flavors"—are sufficient to generate a rejection.[18]

Health Claims and the "NTX" Litigation: Court dockets and administrative petitions uploaded to these compliance blogs yield massive, unredacted troves of rejection data. A high-profile extraction example involves Bellion Vodka, which submitted two label applications (TTB IDs 1425001000403 and 14227001000056) featuring the prominent phrase "Infused with NTX".[19] The TTB issued immediate "Needs Correction" notifications, explicitly rejecting the labels on the grounds that the acronym "NTX" constituted a misleading health claim.[19] The TTB argued that the term implicitly conveyed liver protection and DNA preservation properties to consumers, which violates prohibitions against unsubstantiated health claims on alcoholic beverages.[19] The associated legal filings, commonly indexed as Bellion-Spirits-Appendix-and-supporting-documentation.pdf on legal repositories, expose the exact internal TTB rejection correspondence alongside the unapproved label proofs, providing an immaculate dataset for analysts tracking health-claim enforcement.[19]

The Absinthe Prohibition and Thujone Restrictions: Historical regulatory shifts also create massive clusters of rejected applications. The Bevlog details the TTB's historical policy of issuing blanket rejections for any application referencing the term "absinthe" or depicting related imagery, citing concerns over drug references, hallucinogenic properties, and consumer deception.[21] Following the issuance of Industry Circular 2007-05, the TTB established a strict parameter for approval: any label bearing the term "absinthe" must be accompanied by laboratory data proving the product is "thujone-free".[22] Specifically, the product must undergo a liquid-liquid extraction Gas Chromatography-Mass Spectrometry (GC/MS) test at the Beverage Alcohol Laboratory, verifying that thujone levels are below 10 parts per million.[21] Labels failing to provide this specific supporting documentation are automatically rejected.

Geographical Protests and Surrendered COLAs: Sovos ShipCompliant (sovos.com/shipcompliant/), a major provider of Direct-to-Consumer (DtC) shipping software, extensively tracks the regulatory fallout of labeling errors across the three-tier system.[23] Cross-referencing their whitepapers with wine industry publications reveals specific instances of label surrender forced by political pressure. A notable case study involves the producer *Copper Cane*. The TTB initially approved several COLAs for Copper Cane's Elouan Pinot Noir. However, the producer voluntarily "surrendered" nine specific COLAs following intense protests and legal threats from Oregon state representatives and regional vintners, who argued the labels utilized misleading geographical indicators implying the grapes were sourced from the

Willamette Valley AVA.[25]

Extracting TTB IDs from Industry Troubleshooting Forums

Industry message boards such as *ProBrewer*, the *American Distilling Institute (ADI) Forums*, and *WineBusiness* operate as decentralized, peer-to-peer compliance troubleshooting networks. The sub-forums dedicated to federal regulations are littered with raw, unredacted application data.

To harvest this information, OSINT investigators can execute highly targeted Google dorks against these domains, utilizing operators such as:

site:adiforums.com "needs correction" OR "rejected" OR "TTB ID"

site:probrewer.com "COLA returned" OR "label denial"

The data yield from these queries is exceptionally high. Industry members frequently copy and paste the exact, verbatim text of their TTB rejection emails into forum threads to crowdsource advice. For example, a user operating on the ADI Forums under the handle "TexCF" posted a detailed account of receiving contradictory "Needs Correction" notices from different TTB specialists regarding a blended bourbon and rye product.[26] These threads routinely contain external links to image hosting sites (e.g., Imgur, Photobucket, or native forum attachments) containing the high-resolution files of the rejected labels. This provides the exact "what was submitted" versus "why it was rejected" context required by compliance analysts to build training datasets.[26]

Part 4: Analyzing TTB Official Guidance, Training Materials, and FOIA Repositories

To systematically comprehend the algorithmic constraints and human review protocols that trigger a label rejection, analysts must scrape TTB.gov for official agent training decks, industry Boot Camp webinar slides, and Freedom of Information Act (FOIA) reading room releases. These primary-source documents provide the most authoritative, visual examples of non-compliant formatting available in the public domain.

TTB Boot Camp Webinar Presentations

The TTB regularly publishes internal-grade training materials under its "Boot Camp" series, providing targeted educational tracks for wineries, brewers, and distillers.[27] These presentations, hosted at ttb.gov/about-ttb/outreach/presentations and available as downloadable PDF slide decks, explicitly outline the exact typographical and structural errors that lead to immediate application rejection.[27] Analysis of these decks reveals a rigid taxonomy of compliance failure.

Typology 1: The Government Warning Formatting Trap

The most frequent cause of an automatic "Needs Correction" or "Rejected" status is a micro-deviation from the statutorily mandated Government Warning.[29] According to TTB Boot Camp training slides, any beverage containing at least 0.5% alcohol by volume (ABV) must

feature the warning.[29] The TTB rejects labels based on highly specific formatting parameters:

- **Typography and Bolding:** The introductory words **"GOVERNMENT WARNING"** must be entirely capitalized and rendered in bold type.[27] Failure to apply bolding, or bolding the entire warning statement, results in an immediate rejection.
- **Punctuation and Capitalization:** The subsequent text must be an exact, character-for-character reproduction of the federal statute. Common rejection triggers cited in the training decks include failing to capitalize the "S" and "G" in "Surgeon General," or missing the mandated comma after the word "machinery".[27]
- **Contrast, Placement, and Sizing:** The warning must be readily legible under ordinary conditions, appear on a contrasting background, and remain completely separate and apart from all other label text.[27] Furthermore, type size is strictly regulated based on container volume (e.g., minimum 2 mm for containers larger than 187 ml, and 1 mm for smaller containers).[31]

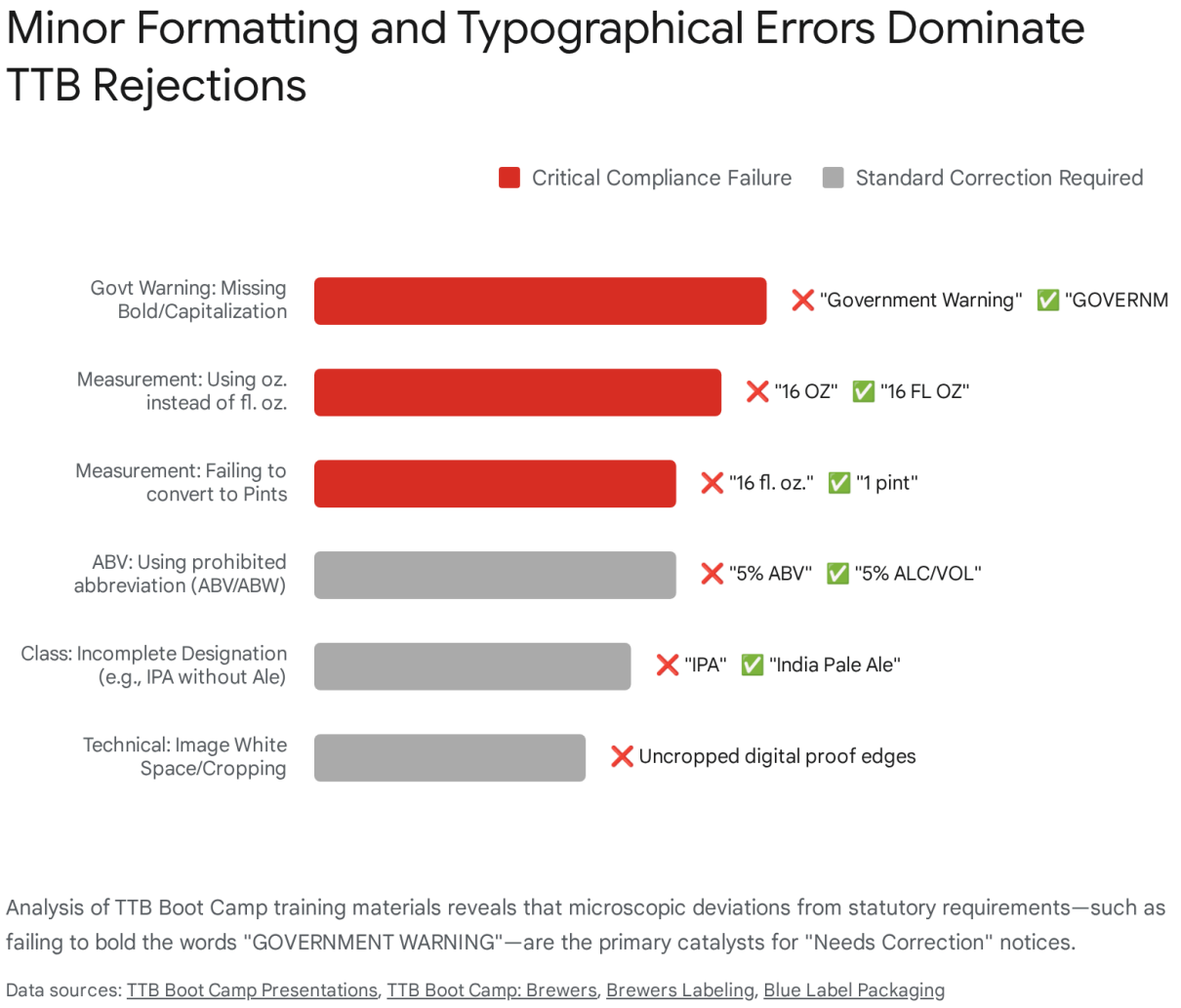
Typology 2: Net Contents and Imperial Measurement Errors

The TTB enforces strict, legacy English measurement standards, particularly concerning malt beverages, resulting in a high volume of application returns.[27]

- **Unit Prohibitions:** Labels that utilize the abbreviation "oz." instead of the mandatory "fl. oz." are automatically flagged for correction.[27]
- **Conversion Failures:** The regulations dictate specific conversion thresholds. A label displaying "16 fl. oz." will be rejected; it must read "1 pint".[27] Similarly, a label reading "22 fl. oz." is non-compliant and must be explicitly converted and displayed as "1 pint 6 fl. oz.".[27]
- **Metric Exclusivity:** While dual measurements are permitted, displaying *only* metric units (e.g., 750 mL) without the required English equivalent is a common trigger for rejection, particularly among international importers.[27]

Typology 3: Class, Type, and ABV Designation Conflicts

- **Alcohol Content Formatting:** Abbreviations indicating alcohol content are strictly controlled. The commonly used acronyms "ABV" (Alcohol by Volume) and "ABW" (Alcohol by Weight) are entirely prohibited by the TTB.[27] Labels must either spell out the words fully or utilize the approved abbreviation "ALC/VOL" (e.g., "5% ALC/VOL"). Submitting a label image featuring the text "5% ABV" guarantees placement in the "Needs Correction" queue.[27]
- **Class/Type Omissions:** On beer labels, recognized style names such as "IPA," "Dunkelweizen," "Bock," or "Tripel" are deemed insufficient as standalone class designations.[27] They must be followed by the word "Ale" or "Beer" (e.g., "India Pale Ale").[27] Furthermore, utilizing the word "Beer:" accompanied by a colon is interpreted by TTB reviewing specialists as a formatting heading rather than a formal class designation, triggering a rejection.[27] For wine, failing to include an appellation of origin when the label features a vintage date or a specific grape varietal represents a critical failure.[27]



Typology 4: Image File and Resolution Rejections

Internal presentation data from COLAs Online Workshops confirms that a significant percentage of rejections are purely technical in nature, relating directly to file formatting rather than legal compliance.

- **White Space and Cropping Constraints:** Label image files uploaded to the COLAs Online portal must be tightly and precisely cropped. Any extraneous white space, background elements, or printer's proof marks remaining on the submitted JPEG or PNG file will cause the image to skew and distort when the system generates the final printable COLA document.[27] This distortion results in an automatic administrative rejection.[27]
- **Resolution and Format Mandates:** The TTB strictly bans the submission of PDF documents for label imagery; only JPEG (.jpg,.jpeg) and PNG formats are accepted, and files cannot exceed 1.5MB.[27] Images must be uploaded at exactly 150 DPI, utilizing an

RGB color mode and medium compression (approximately 70 out of 100).[8] If a TTB reviewing specialist cannot clearly read the microscopic text of a label due to pixelation or compression artifacting, the application is summarily returned for correction.[8]

Exploiting the FOIA Reading Room for Bulk Rejection Data

The TTB Freedom of Information Act (FOIA) Electronic Reading Room serves as the definitive federal source for bulk data regarding historical label rejections.[35] Under subsection (a)(2) of the FOIA, the TTB is statutorily required to make records that were previously disclosed in response to a FOIA request publicly available if the agency determines they are likely to become the subject of subsequent requests.[36]

Analyzing FOIA Logs for Query Construction

Data miners seeking to extract bulk rejections should not blindly submit FOIA requests. Instead, a more efficient methodology involves analyzing the TTB's historical FOIA logs, which are uploaded annually in PDF format (e.g., TTB_FOIA_Log_for_FY_2025.pdf) to ttb.gov/public-information/foia/foia-logs.[37] These logs detail the exact phrasing and legal parameters utilized by industry competitors and investigative journalists to successfully extract hidden rejection data.

For instance, an analysis of the FY2025 FOIA log reveals Request 2025-TTBF-00065, wherein a requester successfully targeted a specific corporate brand's failure history using the following exact verbiage: *"The requester is seeking any and all documents related to labeling of Everclear, distilled spirits, including but not limited to the original labeling application(s), violations or rejections, and approvals, from 2015 until present"*.[38] Another entry, Request 2025-TTBF-00072, demonstrates how to request the complete internal file for a specific application by citing the exact identifier: *"The requester is seeking a copy of TTB COLA ID 02039002000003 under brand name SPARKS VINEYARD & WINERY"*.[38] Replicating this syntax guarantees a higher probability of success when requesting data dumps of "Needs Correction" communications.

Industry Circulars as Precedent Frameworks

The FOIA reading room also hosts a comprehensive archive of historical "Industry Circulars." These documents operate as internal memorandums that dictate massive, industry-wide rejection events. Analyzing these circulars provides the legal framework explaining *why* specific cohorts of labels were systematically denied.

- **Industry Circular 2007-05 (Absinthe):** As previously noted, this circular documents the agency's blanket rejection protocol for applications referencing "absinthe," establishing the "thujone-free" testing mandate required before a COLA could proceed to approval.[21]
- **Industry Circular 2006-01 (Semi-Generic Names):** This document details the strict rejection parameters established for U.S. domestic wines attempting to utilize protected European Union geographical indications (e.g., Retsina, Champagne, Burgundy, Chablis) following the implementation of the 2006 US/EU Wine Trade Agreement.[39]

By cross-referencing the publication dates of these highly disruptive Industry Circulars against

the Public COLA Registry using the "Surrendered" and "Revoked" status filters, a data miner can rapidly generate highly accurate datasets of labels that fell out of compliance due to sudden regulatory shifts.

Part 5: Open-Source Datasets and Algorithmic Label Verification Models

The proliferation of complex, hyper-specific TTB regulations, combined with the severe delays associated with application rejections, has spawned a nascent ecosystem of open-source datasets and machine-learning tools. Hosted across repositories like GitHub and Kaggle, these tools are engineered specifically to scrape bulk COLA data, bypass the TTB's web interfaces, and automate compliance verification via advanced Optical Character Recognition (OCR).

Open-Source Scraping Infrastructure and Data Access

For bulk extraction, the most comprehensive publicly available access point is the **COLA Cloud CLI**, hosted on GitHub at cola-cloud-us/colacloud-cli.[40] This Python-based command-line interface provides programmatic REST API access to over 2.9 million alcohol product label records dating back to 2005, effectively bypassing the slow, paginated TTB Advanced Search web interface.[40]

- **Execution Protocol:** Analysts can install the tool via the standard package installer: pip install colacloud-cli.[40]
- **Granular Query Capabilities:** The CLI allows for highly granular filtering that exceeds the capabilities of the public registry, outputting massive .json or .csv files for direct database ingestion. A standard execution command to extract data for a specific commodity and ABV range is: cola colas list --product-type "distilled spirits" --abv-min 40 --abv-max 50 --json | jq '.data.brand_name'.[40] The system also utilizes LLM-powered enrichment to extract descriptive fields such as ingredients and tasting notes directly from the imagery.[40]
- **Kaggle Dataset Integrations:** For data scientists seeking pre-compiled structured data without utilizing the CLI, a static sample of this dataset is hosted on Kaggle under the repository colacloud/ttb-colas-demo, providing .csv files of application forms mapped to their label imagery.[41] Other supplementary datasets on Kaggle include the jessemostipak/beer-production dataset (a TidyTuesday set offering historical context on barrel production and removal volumes) [42], and the axlifreeway/retail-alcohol-detection-dataset, which provides thousands of images of alcohol products on retail shelves designed for training object detection models in cluttered environments.[43]

AI-Powered OCR and Compliance Verification Repositories

Because ALFD specialists routinely reject labels for microscopic typographical errors—such as a missing comma in a warning statement or an improper fluid ounce conversion—developers have begun open-sourcing AI tools designed to algorithmically pre-screen labels before formal

submission.

The GitHub repository ArkieCoder/ttb-verifier represents the bleeding edge of this specific OSINT and compliance vector.[44] Engineered specifically as an AI-powered TTB label verification tool for compliance agents, it utilizes the Streamlit framework and the EasyOCR library to automate the inspection process.[44]

- **Operational Mechanism:** The user uploads a batch of proposed high-resolution label images. The Python script executes an OCR pass across the image to extract all readable text blocks.[40]
- **Algorithmic Verification:** The tool utilizes RapidFuzz (a high-speed string-matching library) to compare the extracted text against the hardcoded application data and TTB statutory requirements.[44] It specifically verifies the "Brand Name," confirms the precise presence and formatting of "ALC/VOL," and runs a character-by-character string match against the mandated "GOVERNMENT WARNING".[44]
- **Output Generation:** Upon completion, the script generates a color-coded checklist with match, review, and mismatch verdicts accompanied by confidence scores.[44] This programmatic output effectively replicates the TTB's internal "Needs Correction" validation process, allowing producers to catch fatal errors before waiting weeks for a human specialist's review.

Other specialized repositories address specific challenges within the alcohol OCR domain. For example, AntoninLeroy/wine_label_reader_toolkit provides custom computer vision pipelines designed specifically for reading wine labels, highlighting the algorithmic difficulty of extracting stylized, cursive, and curved text from cylindrical bottles using standard, off-the-shelf OCR libraries.[45] Similarly, Sid2697/Beer-Label-Classification utilizes Scale-Invariant Feature Transform (SIFT) and Oriented FAST and Rotated BRIEF (ORB) algorithms to classify beer bottle labels against a clean database of imagery.[46] Furthermore, repositories like iSE-UET-VNU/Cola provide data-centric tools that leverage local neighborhood similarities to detect and eliminate corrupted labels within massive real-world datasets, thereby improving the accuracy of downstream Deep Learning models.[47]

Open-Source Infrastructure for TTB COLA Extraction and Analysis

Repo/Tool	Platform	Tech Stack	Core Functionality
colacloud-cli	GitHub / REST API	Python	Command-line tool to access and search the TTB COLA registry of 2.9M+ records. Supports full-text search, filtering (ABV, origin, date, brand, product type), and JSON/CSV export for bulk extraction. Integrates LLM enrichment and barcode extraction.
my ttb sem1	GitHub / Hugging Face Spaces	Python, Streamlit, EasyOCR, RapidFuzz	AI-powered TTB label verification tool for compliance agents. Automates comparison of brand name, ABV, and government warnings against application data, generating color-coded match/review/mismatch verdicts.
ttb-verifier	GitHub	Python, FastAPI, Docker, Terraform	AI-powered alcohol beverage label verification system engineered specifically for U.S. Treasury Department TTB compliance checks.
wine_label_reader_toolkit	GitHub	Python (OCR Libraries)	Simplifies the extraction of text from standard photographs of wine bottles using Optical Character Recognition (OCR) to read complex label designs.
bonded-still	GitHub	Shell	Supply chain SaaS application focusing on distillery compliance, TTB excise tax automation, and bonded warehouse management.
cidery-manager	GitHub	PHP (Laravel)	Open-source record management system designed specifically to handle operations and TTB compliance tracking for wineries and cideries.
CS3733-COLA	GitHub	Java	Academic prototype of the TTB COLAs system, created as coursework to simulate alcoholic beverage application workflows.

Developers are increasingly leveraging Python-based OCR pipelines and REST APIs to bypass the TTB's manual interfaces, automating both the extraction of approved labels and the detection of non-compliant formatting.

Data sources: COLA Cloud, GitHub (TTB Topics), GitHub (colacloud-cli), GitHub (wine_label_reader_toolkit)

Conclusion: Strategic Directives for Compliance Extraction

The pursuit of non-compliant, rejected, or corrected TTB COLA data requires a highly asymmetrical analytical approach. Because the TTB actively and successfully safeguards "Needs Correction" and "Rejected" applications behind an authenticated firewall to protect the commercial interests of industry members during product development, direct querying of these specific failure statuses via the Public COLA Registry is technologically impossible.[5]
To build exhaustive compliance datasets and train accurate verification models, data miners must systematically circumvent the central registry's limitations. This is achieved through a multi-vector strategy: scraping the Public Registry for "Surrendered" and "Revoked" anomalies that serve as proxies for post-market rejections [5]; mining legal blogs and specialized trade forums for unredacted rejection notices, leaked TTB IDs, and side-by-side visual comparisons of failed label proofs [18]; and conducting deep textual analysis of the TTB's own internal training materials and FOIA logs to map the precise typographical, formatting, and measurement triggers that cause applications to fail during human review.[27] By fusing this historically mined, qualitative data with emerging open-source OCR tools, REST APIs, and string-matching libraries [40], investigators can effectively reverse-engineer the TTB's compliance algorithms. This synthesis allows for the creation of massive shadow databases of labeling failures, enabling automated compliance verification without ever needing to breach the agency's authenticated perimeter.

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